Introduction to Data Vault Modeling



Kent Graziano
Data Vault Master and Oracle ACE
TrueBridge Resources

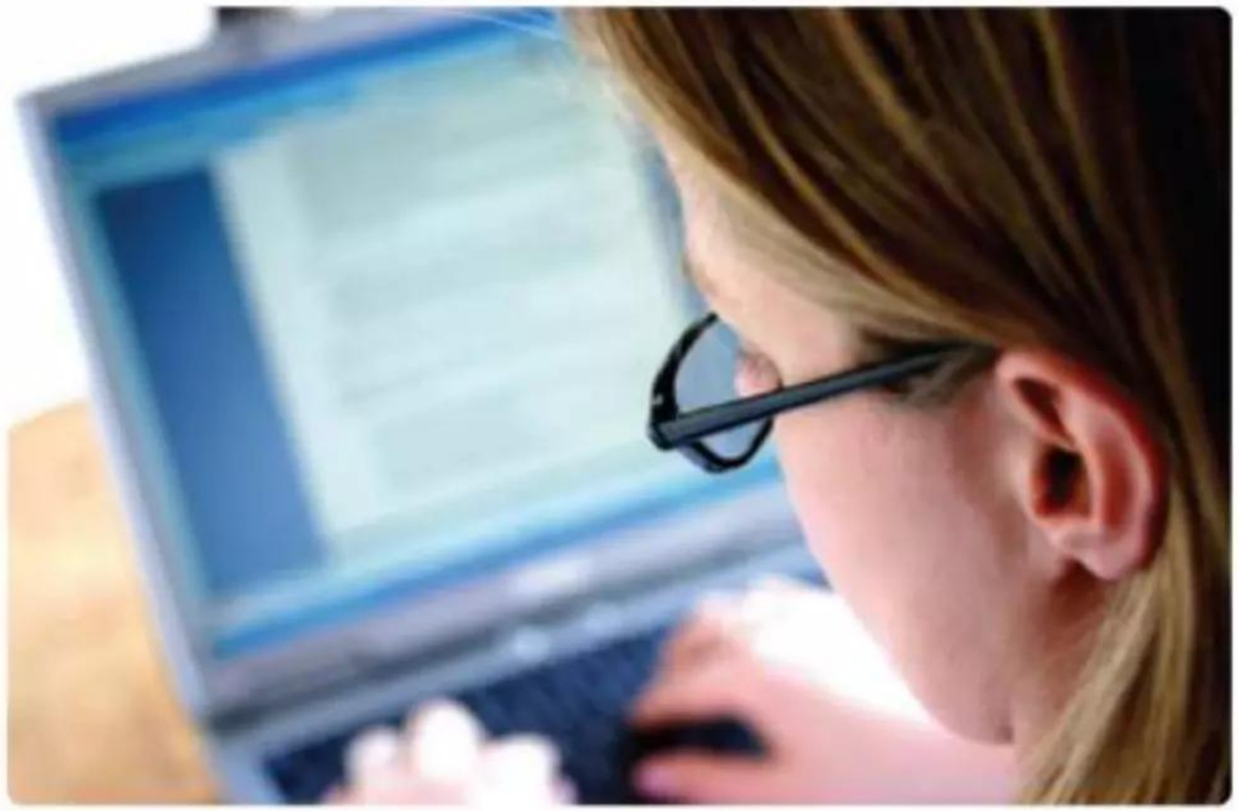
OOW 2011
Session #05923

My Bio

- Kent Graziano
  - Certified Data Vault Master
  - Oracle ACE (BI/DW)
  - Data Architecture and Data Warehouse Specialist
    - 30 years in IT
    - 20 years of Oracle-related work
    - 15+ years of data warehousing experience
  - Co-Author of
    - The Business of Data Vault Modeling (2008)
    - The Data Model Resource Book (1st Edition)
    - Oracle Designer: A Template for Developing an Enterprise Standards Document
  - Past-President of Oracle Development Tools User Group (ODTUG) and Rocky Mountain Oracle User Group
  - Co-Chair BIDW SIG for ODTUG

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Oracle Development Tools User Group

www.odtug.com

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What Is a Data Warehouse?

"A subject-oriented, integrated, time-variant, non-volatile collection of data in support of management's decision making process."

W.H. Inmon

"The data warehouse is where we *publish used data*."

Ralph Kimball

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Inmon's Definition

- Subject oriented
  - Developed around logical data groupings (subject areas) not business functions
- Integrated
  - Common definitions and formats from multiple systems
- Time-variant
  - Contains historical view of data
- Non-volatile
  - Does not change over time
  - No updates

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Data Vault Definition

The Data Vault is a detail oriented, historical tracking and uniquely linked set of normalized tables that support one or more functional areas of business.

It is a hybrid approach encompassing the best of breed between 3rd normal form (3NF) and star schema. The design is flexible, scalable, consistent, and adaptable to the needs of the enterprise. It is a data model that is architected specifically to meet the needs of today's enterprise data warehouses.

Dan Linstedt: Defining the Data Vault TDAN.com Article

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Why Bother With Something New?

Old Chinese proverb:
'Unless you change direction, you're apt to end up where you're headed.'



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Why do we need it?

- We have seen issues in constructing (and managing) an ***enterprise data warehouse model*** using 3rd normal form, or Star Schema.
  - 3NF – Complex PKs when cascading snapshot dates (time-driven PKs)
  - Star – difficult to re-engineer fact tables for granularity changes
- These issues lead to break downs in flexibility, adaptability, and even scalability

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Data Vault Time Line

E.F. Codd invented relational modeling

Chris Date and Hugh Darwen Maintained and Refined Modeling

1976 Dr Peter Chen Created E-R Diagramming

Mid 70's AC Nielsen Popularized Dimension & Fact Terms

1990 – Dan Linstedt Begins R&D on Data Vault Modeling

1960 1970 1980 1990 2000

Early 70's Bill Inmon Began Discussing Data Warehousing

Mid 60's ***Dimension & Fact*** Modeling presented by General Mills and Dartmouth University

Late 80's – Barry Devlin and Dr Kimball Release "Business Data Warehouse"

Mid 80's Bill Inmon Popularizes Data Warehousing

Mid – Late 80's Dr Kimball Popularizes Star Schema

2000 – Dan Linstedt releases first 5 articles on Data Vault Modeling

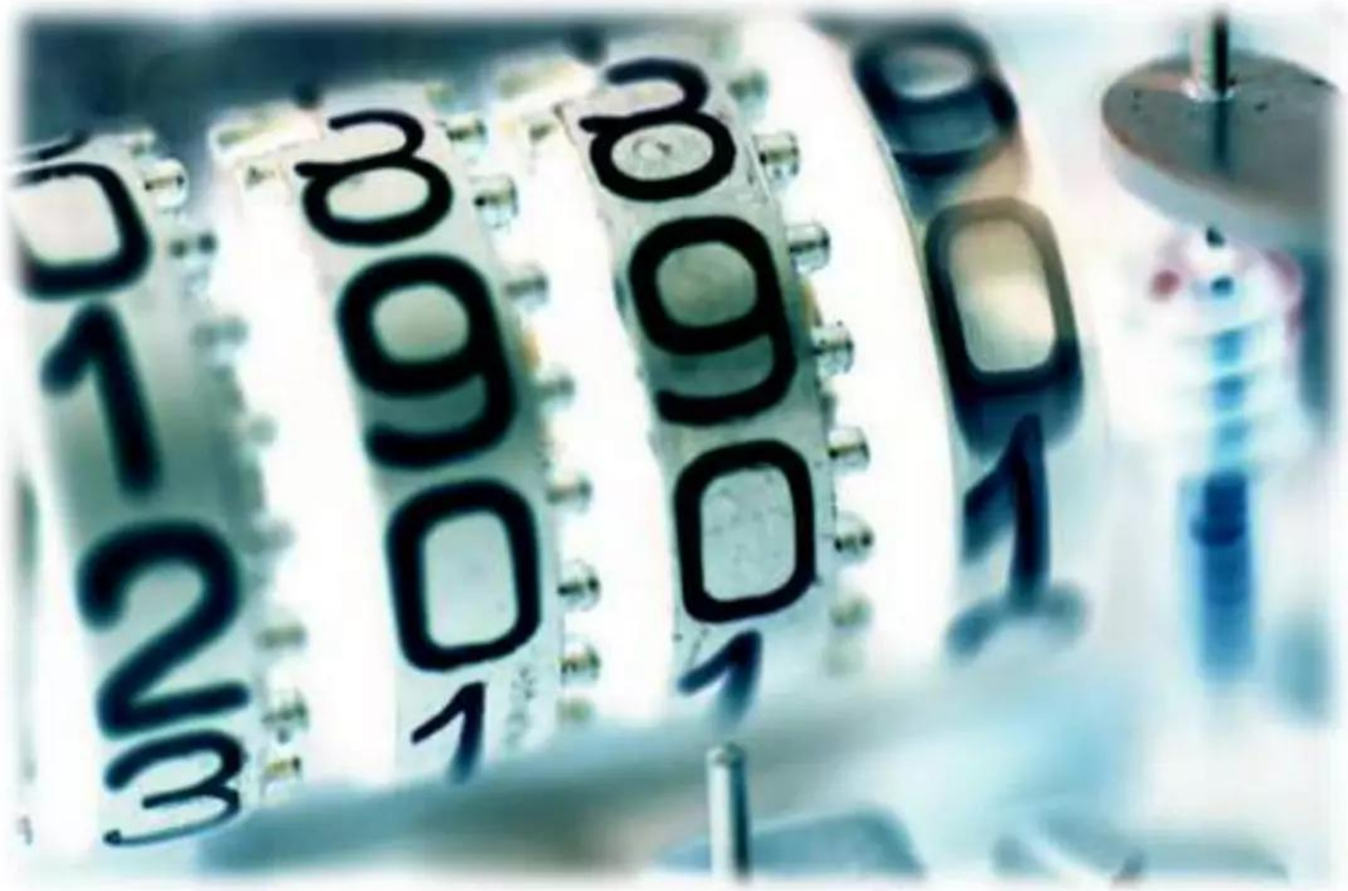
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Data Vault Evolution

- The work on the Data Vault approach began in the early 1990s, and completed around 1999.
- Throughout 1999, 2000, and 2001, the Data Vault design was tested, refined, and deployed into specific customer sites.
- In 2002, the industry thought leaders were asked to review the architecture.
  - This is when I attend my first DV seminar in Denver and met Dan!
- In 2003, Dan began teaching the modeling techniques to the mass public.

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Data Vault Modeling…



Took 10 years of Research and Design, including TESTING to become flexible, consistent, and scalable

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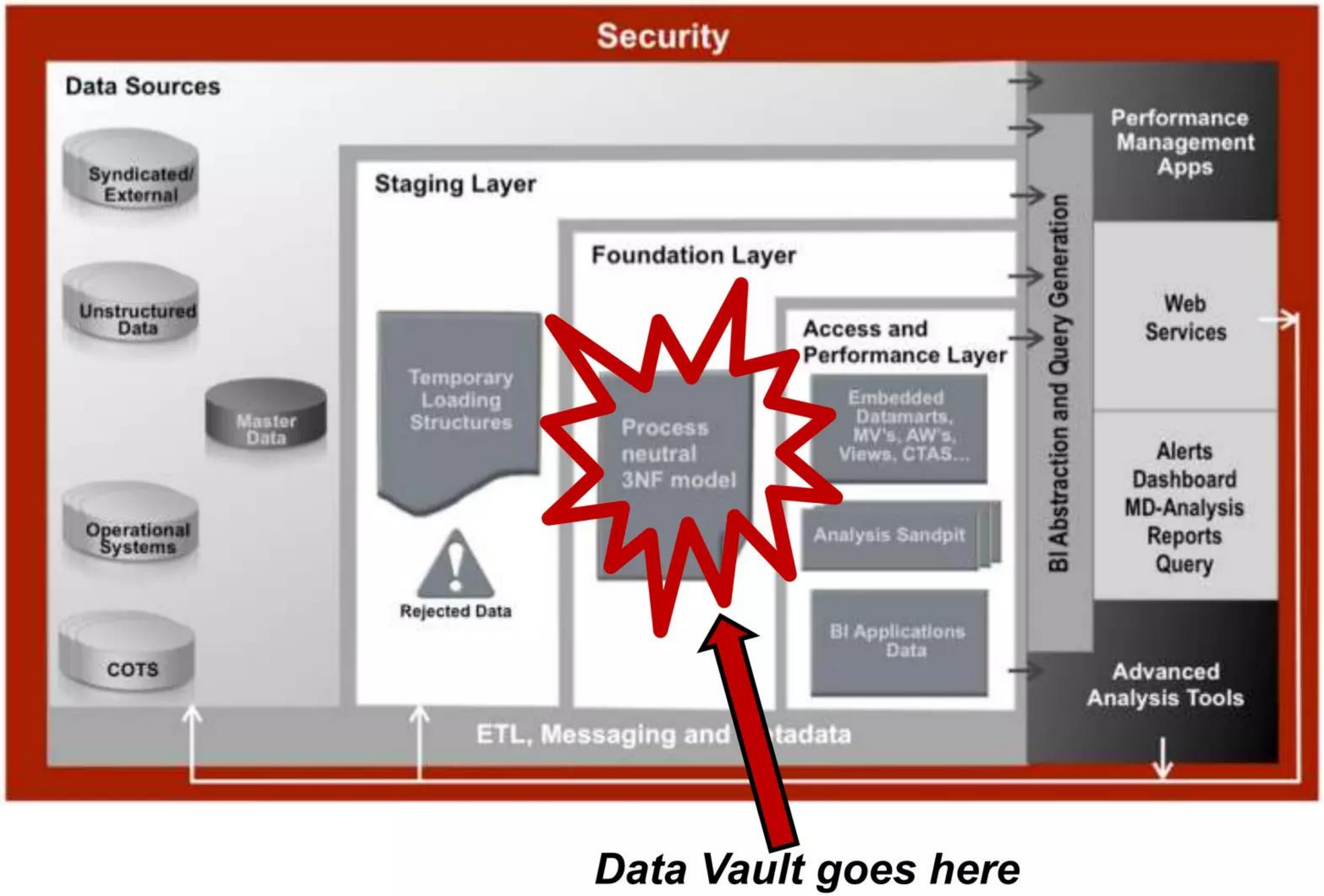
Where does a Data Vault Fit?



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Where does a Data Vault Fit?

Oracle's Next Generation Data Warehouse Reference Architecture



(C) Oracle Corp

EDW
DATA VAULT

3 Simple Structures

1 Hub

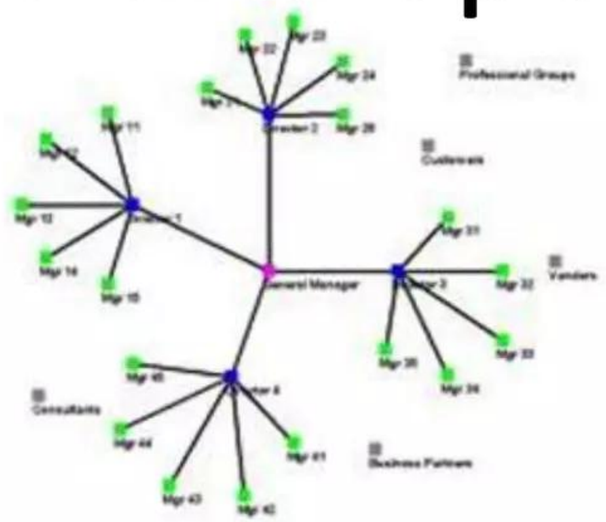
2 Link

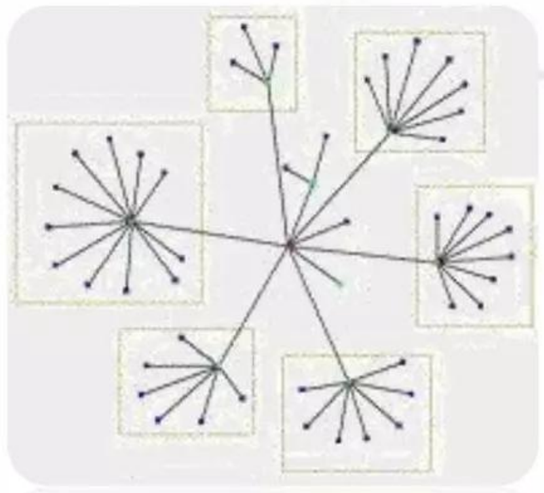
3 Satellite

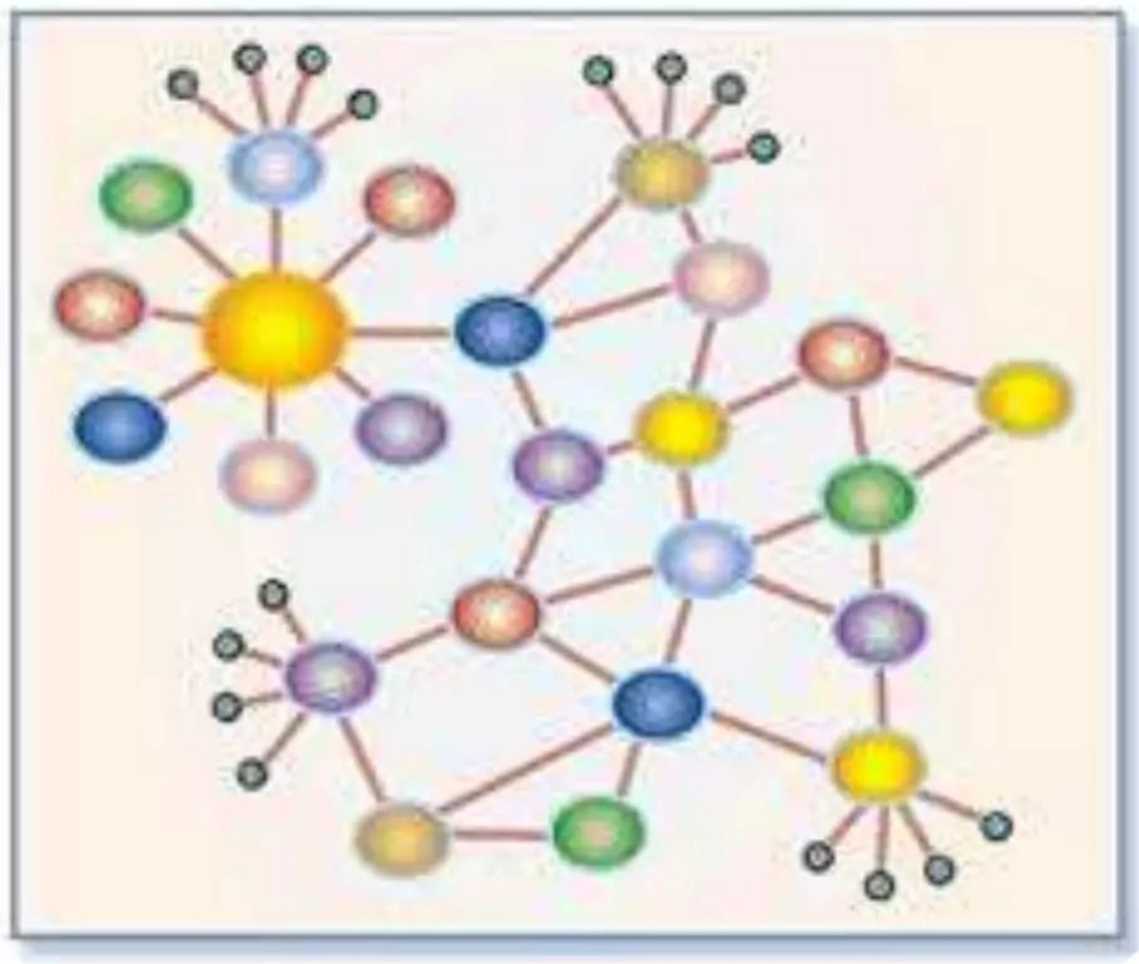
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Hub and Spoke = Scalability









http://www.nature.com/ng/journal/v29/n2/full/ng1001-105.html

If nature uses Hub & Spoke, why shouldn't we?
Genetics scale to billions of cells,
the Data Vault scales to Billions of records



Hubs = Neurons

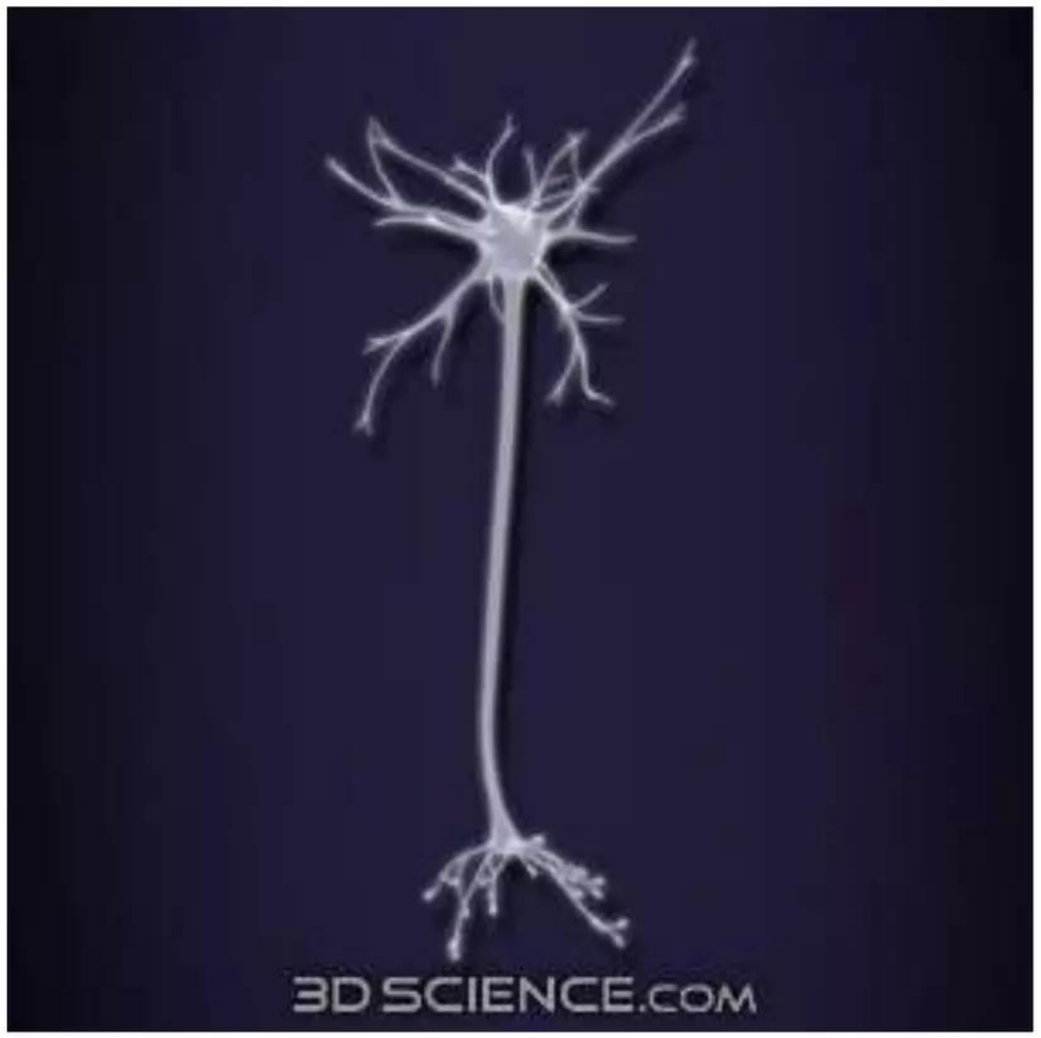


Very similar to a neural network,
The Hubs create the base structure

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Links = Dendrite + Synapse



In neural networks,
Dendrites & Synapses fire to pass messages,
The Links dictate associations, connections

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Satellites = Memories



Perception, understanding and processing
These all ***describe*** the memory
Satellites house descriptors that can change over time

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National Drug Codes + Orange Book of Drug Patent Applications

A WORKING EXAMPLE

http://www.accessdata.fda.gov/scripts/cder/ndc/default.cfm
http://www.fda.gov/Drugs/InformationOnDrugs/ucm129662.htm

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1. Hub = Business Keys



Drug Label Code

Product Number

NDA Application #

Firm Name

Dose Form Code

Drug Listing

Patent Number

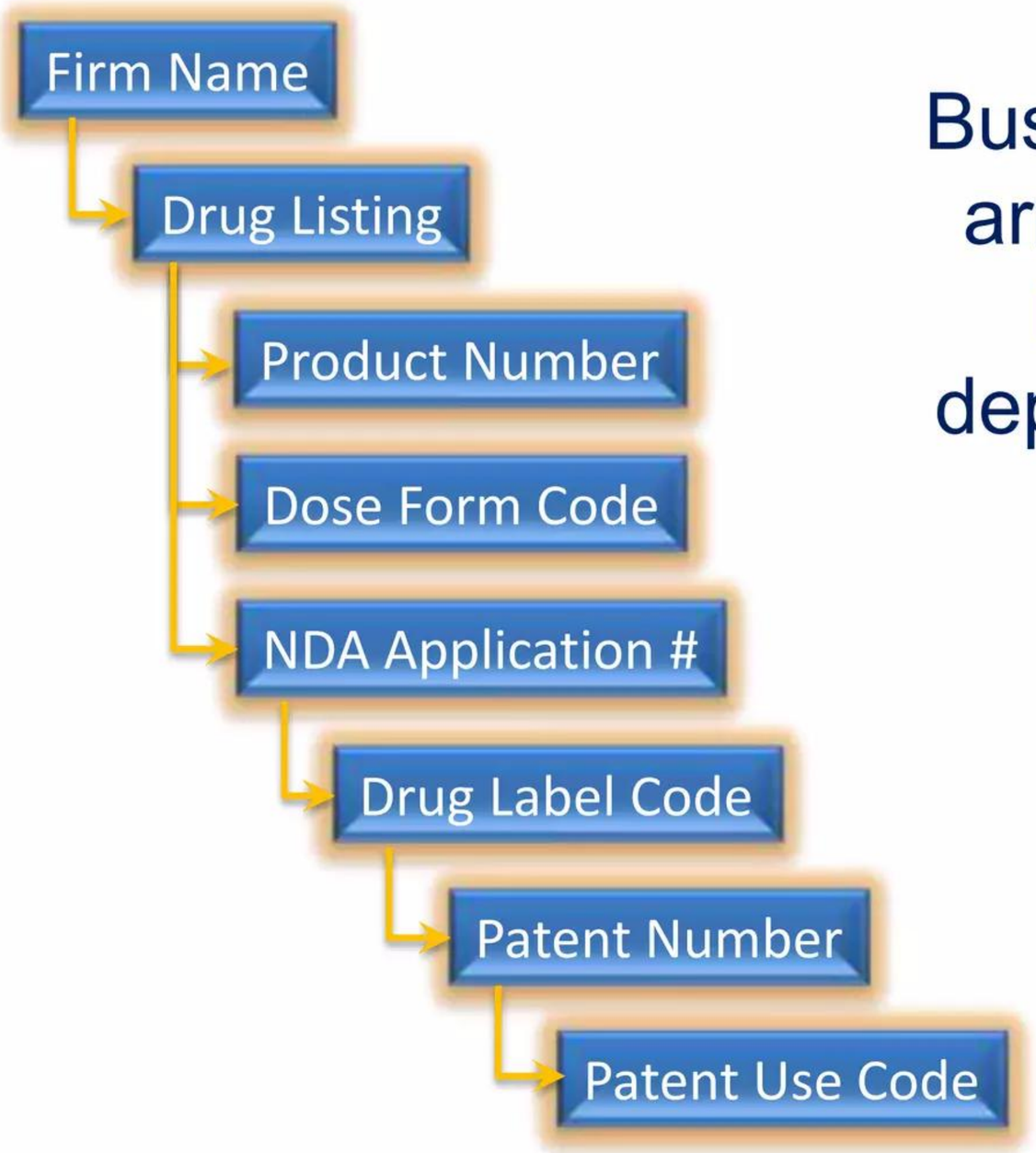
Patent Use Code

Hubs = Unique Lists of Business Keys
Business Keys are used to
TRACK and IDENTIFY key information

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Business Keys = Ontology





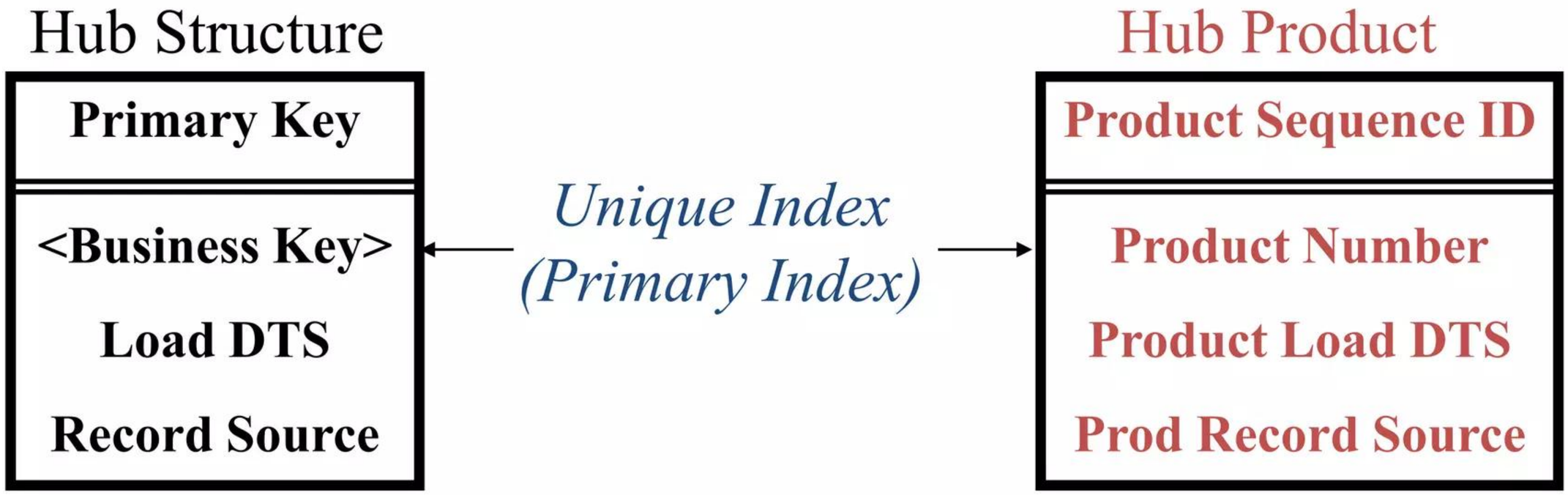
Business Keys ***should*** be arranged in an ontology
In order to learn the dependencies of the data set

NOTE: Different Ontologies represent different views of the data!

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Hub Entity

A Hub is a list of unique business keys.



Note:

- A Hub's Business Key is a unique index.
- A Hub's Load Date represents the FIRST TIME the EDW saw the data.
- A Hub's Record Source represents: First – the "Master" data source (on collisions), if not available, it holds the origination source of the actual key.

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Business Keys

- What exactly are Business Keys?
  - Example 1:
    - Siebel has a “system generated” customer key
    - Oracle Financials has a “system generated” customer key
    - ***These are not business keys***. These are keys used by each respective system to track records.
  - Example 2:
    - Siebel Tracks customer name, and address as unique elements.
    - Oracle Financials tracks name, and address as unique elements.
    - These *are* business keys.
- What we want in the hub, are sets of natural business keys that uniquely identify the data – across systems.
- Stay away from “system generated” keys if possible.
  - System Generated keys will cause damage in the integration cycle if they are *not unique* across the enterprise.

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Hub Definition

- What Makes a Hub Key?
  - A Hub is based on an *identifiable* business key.
  - An identifiable business key is an attribute that is used in the source systems to locate data.
  - The business key has a very low propensity to change, and usually is not editable on the source systems.
  - The business key has the *same semantic meaning*, and the *same granularity* across the company, but not necessarily the same format.
- Attributes and Ordering
  - All attributes are mandatory.
  - Sequence ID 1st, Busn. Key 2nd , Load Date 3rd ,Record Source Last (4th).
  - All attributes in the Business Key form a UNIQUE Index.

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The *technical* objective of the Hub is to:

- Uniquely list all possible business keys, good, bad, or indifferent of where they originated.
- Tie the business keys in a 1:1 ratio with surrogate keys (giving meaning to the surrogate generated sequences).
- Provide a consolidation and attribution layer for clear horizontal definition of the business functionality.
- Track the arrival of data, the *first* time it appears in the warehouse.
- Provide right-time / real-time systems the ability to load transactions without descriptive data.

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Hub Table Structures

HUB_LBL_CODE

LBLCODE_SQN
LBLCODE LBLCODE_LDTS LBLCODE_RSRC

HUB_PROD_NUM

PROD_NUM_SQN
PROD_NUM PROD_NUM_LDTS PROD_NUM_RSRC

HUB_FIRM_NAME

FIRM_SQN
FIRM_NAME FIRM_LDTS FIRM_RSRC

HUB_NDA_APPLICATION

APPL_SQN
APPL_NUM APPL_LDTS APPL_RSRC

HUB_LISTINGS

LIST_SQN
LISTING_SEQ_NO LIST_LDTS LIST_RSRC

HUB_PATENT_USE_CODE

PAT_USE_CODE_SQN
PAT_USE_CODE PAT_USE_CODE_LDTS PAT_USE_CODE_RSRC

HUB_PATENT_NUMBER

PAT_NUM_SQN
PAT_NUMBER PAT_NUM_LDTS PAT_NUM_RSRC

HUB_DOSE_FORM_CODE

DOSE_FORM_SQN
DOSE_FORM_LDTS DOSE_FORM_CODE DOSE_FORM_RSRC

SQN = Sequence (insertion order)

LDTS = Load Date (when the Warehouse first sees the data)

RSRC = Record Source (System + App where the data ORIGINATED)

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Sample Hub Product

ID	PRODUCT #	LOAD DTS	RCRD SRC
1	MFG-PRD123456	6-1-2000	MANUFACT
2	P1235	6-2-2000	CONTRACTS
3	*P1235	2-15-2001	CONTRACTS
4	MFG-1235	5-17-2001	MANUFACT
5	1235-MFG	7-14-2001	FINANCE
6	1235	10-13-2001	FINANCE
7	PRD128582	4-12-2002	MANUFACT
8	PRD125826	4-12-2002	MANUFACT
9	PRD128256	4-12-2002	MANUFACT
10	PRD929929-*	4-12-2002	MANUFACT

Unique Index

Notes:

- **ID is the surrogate sequence number (Primary Key)**
- **What does the load date tell you?**
- **Do you notice any overloaded uses for the product number?**
- **Are there similar keys from different systems?**
- **Can you spot entry errors?**
- **Are any patterns visually present?**

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2. Links = Associations

Firms Generate Labels

Firms Generate Product Listings

Firms Manufacture Products

Listings Contain Labeler Codes

Listings for Products are in NDA Applications

Links = Transactions and Associations
They are used to hook together multiple sets of information (i.e., Hubs)

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Associations = Ontological Hooks

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Business Keys are associated by many linking factors, these links comprise the associations in the hierarchy.

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Link Definitions

- <u>What Makes a Link?</u>
  - A Link is based on *identifiable* business element *relationships*.
    - Otherwise known as a foreign key,
    - AKA a business event or transaction between business keys,
  - The relationship shouldn't change over time
    - It is established as a *fact* that occurred at a specific point in time and will remain that way forever.
  - The link table may also represent a *hierarchy*.

- <u>Attributes</u>
  - All attributes are mandatory

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Link Entity

A Link is an intersection of business keys.
It can contain Hub Keys and Other Link Keys.

Link Structure

Primary Key
{Hub Surrogate Keys 1..N}
Load DTS
Record Source

Unique Index (Primary Index)

Link Line-Item

Link Line Item Sequence ID
Hub Product Sequence ID
Hub Order Sequence ID
Load DTS
Record Source

Note:

- A Link's Business Key is a Composite Unique Index
- A Link's Load Date represents the FIRST TIME the EDW saw the relationship.
- A Link's Record Source represents: First – the "Master" data source (on collisions), if not available, it holds the origination source of the actual key.

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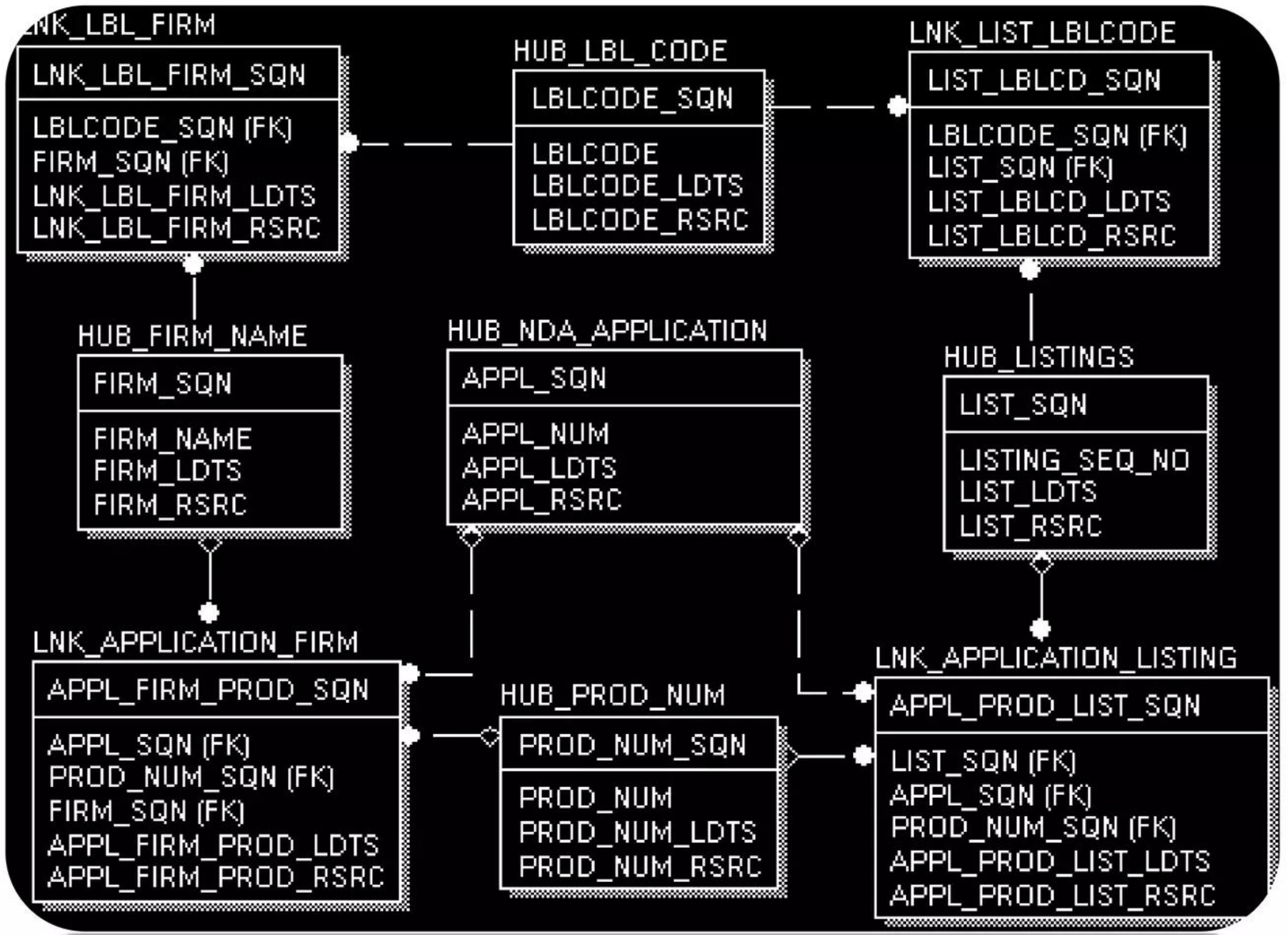
Modeling Links - 1:1 or 1:M?

- Today:
  - Relationship is a 1:1 so why model a Link?
- Tomorrow:
  - The business rule can change to a 1:M.
  - You discover new data later.
- With a Link in the Data Vault:
  - No need to change the EDW structure.
  - Existing data is fine.
  - New data is added.

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Link Table Structures



SQN = Sequence (insertion order)

LDTS = Load Date (when the Warehouse first sees the data)

RSRC = Record Source (System + App where the data ORIGINATED)

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Sample Link Entity - Relationship

Hub Customer

CSID	CUST #	LOAD DTS	RCRD SRC
1	ABC123456	10-12-2000	MFG
2	DKEF	1-25-2001	CONTRACTS

Order Satellite

Hub Order

OrdID	ORDER #	LOAD DTS	RCRD SRC
1	ORD0001	10-12-2000	MFG
2	ORD0002	10-2-2000	CONTRACTS

Link Cust Order

LSEQID	CSID	OrdID	LOAD DTS	RCRD SRC
1000	1	1	10-14-2000	FINANCE
1001	1	2	10-14-2000	FINANCE

Link Order-Details

LSEQID	OrdID	PID	LIT	LOAD DTS	RCRD SRC
1000	1	100	1	10-14-2000	FINANCE
1001	1	101	2	10-14-2000	FINANCE

Order Details Satellite

Product Satellite

Hub Product

PID	PRODUCT #	LOAD DTS	RCRD SRC
100	PRD128582	10-14-2000	MFG
101	PRD128256	10-14-2000	MFG

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Sample Link Entity - Hierarchy

Link Customer Rollup

From CSID	To CSID	LOAD DTS	RCRD SRC
1	NULL	10-14-2000	FINANCE
2	1	10-22-2000	FINANCE
3	1	2-15-2001	FINANCE
4	2	4-3-2001	HR
5	2	6-4-2001	SALES

Hub Customer

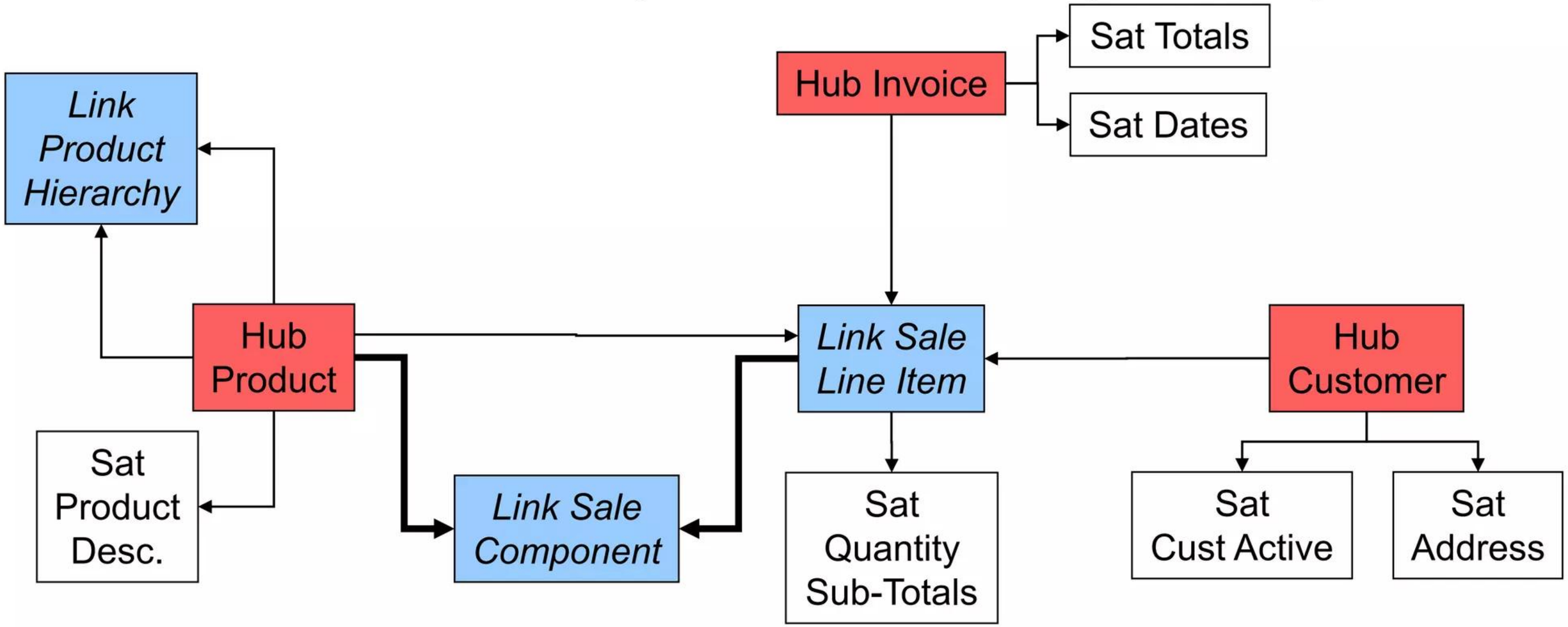
ID	CUSTOMER #	LOAD DTS	RCRD SRC
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-22-2000	CONTRACTS
3	DKEF	1-25-2001	CONTRACTS
4	KKO92854_dd	3-7-2001	CONTRACTS
5	LLOA_82J5J	6-4-2001	SALES
6	HUJI_BFIOQ	8-3-2001	SALES
7	PPRU_3259	2-2-2002	FINANCE
8	PAFJG2895	2-2-2002	CONTRACTS
9	929ABC2985	2-2-2002	CONTRACTS
10	93KFLLA	2-2-2002	CONTRACTS

Note:

- If you have logic – you can roll together customers, or companies, or sub-assemblies, bill of materials, etc..
- We do not want to disturb the *facts* (underlying data in the hub), but we do want to re-arrange hierarchies at different points over time.

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Link To Link (Link Sale Component)



Note:

- Link Sale Component provides a shift in grain.
- Link Sale Component allows for configurable options of products tracked on a single line-item product sold.
- Link Sale Component provides for sub-assembly tracking.

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3. Satellites = Descriptors



Firm Locations

Listing Formulation

Patent Expiration Info

Product Ingredients

Listing Medication Dosages

Drug Packaging Types

Satellites = Descriptors
These data provide ***context*** for the keys (Hubs)
And for the associations (Links)

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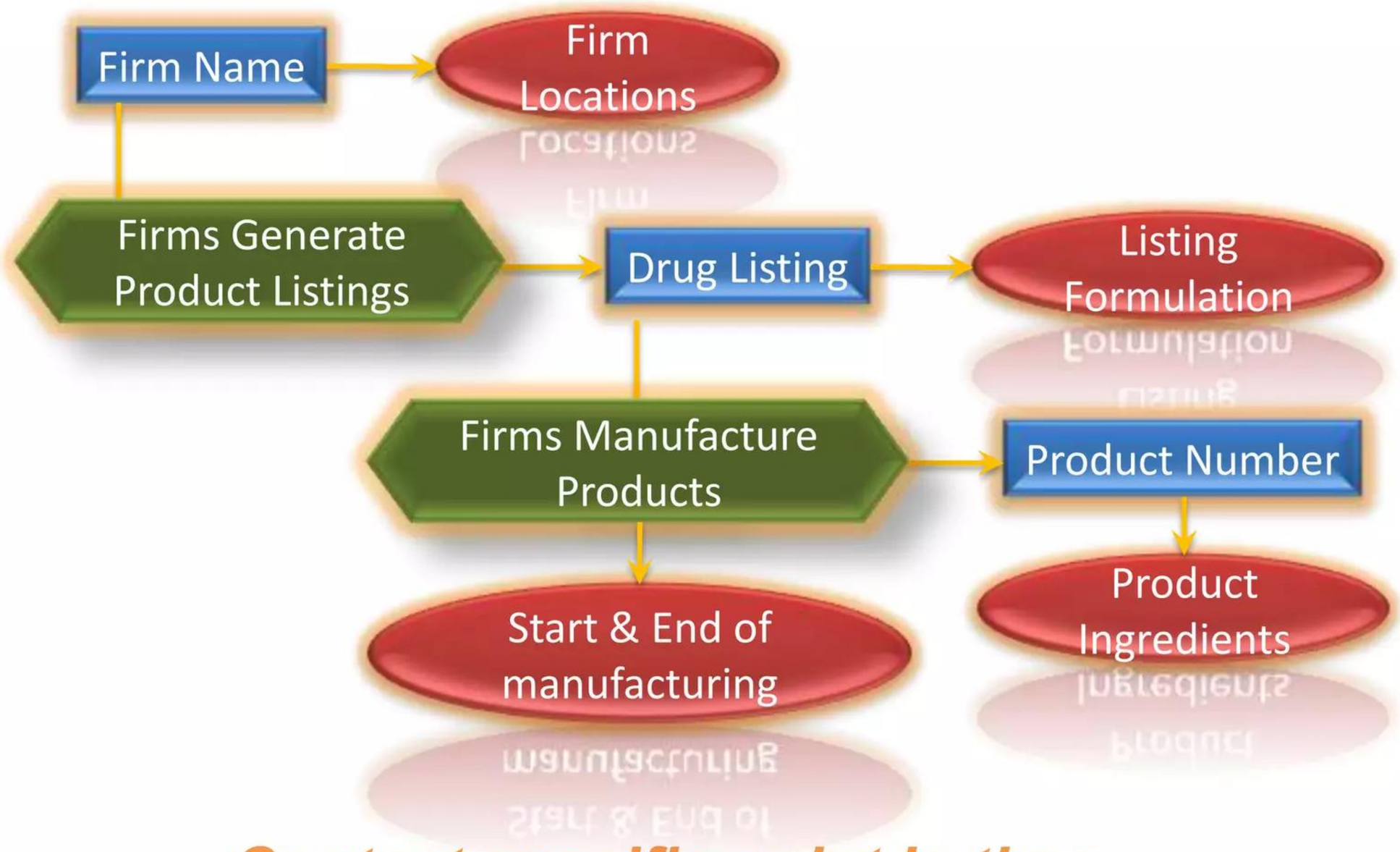
Satellite Definitions

- <u>What Makes a Satellite?</u>
  - A Satellite is based on an *non-identifying* business elements.
    - Attributes that are descriptive data, often in the source systems known as descriptions, or free-form entry, or computed elements.
  - The Satellite data changes, sometimes rapidly, sometimes slowly.
    - The Satellites are separated by *type of information* and *rate of change*.
  - The Satellite is *dependent* on the Hub or Link key as a parent,
    - Satellites are *never* dependent on more than one parent table.
    - The Satellite is *never* a parent table to any other table (no snow flaking).
- <u>Attributes and Ordering</u>
  - All attributes are mandatory – EXCEPT END DATE.
  - Parent ID 1st, Load Date 2nd, Load End Date 3rd,Record Source Last.

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EDW
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Descriptors = Context



Context specific point in time warehousing portion

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Satellite Entity

A Satellite is a time-dimensional table housing detailed information about the Hub's or Link's business keys.

Hub Primary Key Load DTS
Extract DTS *Load End Date* Detail Business Data <Aggregation Data>
{Update User} {Update DTS} Record Source

Customer # Load DTS
Extract DTS *Load End Date* Customer Name Customer Addr1 Customer Addr2
{Update User} {Update DTS} Record Source

- ***Satellites are defined by TYPE of data and RATE OF CHANGE***
- ***Mathematically – this reduces redundancy and decreases storage requirements over time (compared to a Star Schema)***

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Satellite Entity- Details

- A Satellite has only 1 foreign key; it is dependent on the *parent* table (Hub or Link)
- A Satellite may or may not have an *"Item Numbering"* attribute.
- A Satellite's Load Date represents the date *the EDW saw the data* (must be a delta set).
  - This is not Effective Date from the Source!
- A Satellite's Record Source represents the actual source of the row (unit of work).
- To avoid Outer Joins, you must ensure that every satellite has at least 1 entry for every Hub Key.

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Satellite Table Structures

SAT_LBL_FIRMS

LBLCODE_SQN: numeric(10) (FK) FIRM_LDTS: datetime
FIRM_LEDTS: datetime FIRM_NAME: varchar(150) ADDR_HEADER: varchar(40) STREET: varchar(40) PO_BOX: varchar(9) FOREIGN_ADDR: varchar(40) CITY: varchar(30) STATE_CD: varchar(2) ZIP: varchar(9) PROVINCE_NAME: varchar(30) COUNTRY: varchar(40) FIRM_RSRC: varchar(15)

SAT_DOSE_FORM

DOSE_FORM_SQN: numeric(10) (FK) DOSAGE_LDTS: datetime
DOSAGE_LEDTS: datetime DOSAGE_NAME: varchar(240) DOSAGE_RSRC: varchar(15)

SAT_LISTINGS

LIST_SQN: numeric(10) (FK) LIST_LDTS: datetime
LIST_LEDTS: datetime PRODUCT_CD: varchar(4) STRENGTH: varchar(10) UNIT_CODE: varchar(15) RX_OTC: varchar(1) TRADENAME: varchar(100) LIST_RSRC: varchar(15)

SAT_SCHEDULE

LIST_SQN: numeric(10) (FK) SCHDL_SUB_SQN: int SCHDL_LDTS: datetime
SCHEDULE: numeric(1) SCHDL_RSRC: varchar(15)

SAT_PATENT_APPL

PAT_APPL_PROD_SQN: numeric(10) (FK) PAT_LDTS: datetime
PAT_LEDTS: datetime PAT_EXPIRATION_DATE: varchar(29) PAT_DRUG_SUBSTANCE_FLG: char(1) PAT_DRUG_PROD_FLG: char(1) PAT_DELIST_FLG: char(1) PAT_RSRC: varchar(15)

SAT_APPL_LIST_PROD

APPL_FIRM_PROD_SQN: numeric(10) (FK) APPL_PROD_LIST_LDTS: datetime
APPL_PROD_LIST_LEDTS: datetime APPL_INGREDIENT_NAME: varchar(255) APPL_DOSE_FORM_ROUTE: varchar(90) APPL_TRADENAME: varchar(155) APPL_FIRM_NAME: varchar(65) APPL_STRENGTH: varchar(155) APPL_THERAP_EQUIV_CODE: varchar(2) APPL_APPROVAL_DATE: varchar(35) APPL_REF_LISTED_DRUG_CODE: varchar(5) APPL_RX_OTC_TYPE_CODE: varchar(5) APPL_RSRC: varchar(15)

SQN = Sequence (parent identity number)

LDTS = Load Date (when the Warehouse first sees the data)

LEDTS = End of lifecycle for superseded record

RSRC = Record Source (System + App where the data ORIGINATED)

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Satellite Entity – Hub Related

Hub Customer

ID	CUSTOMER #	LOAD DTS	RCRD SRC
0	N/A	10-12-2000	SYSTEM
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-2-2000	CONTRACTS
3	*ABC5525-25*	***10-1-2000***	***FINANCE***

CUSTOMER NAME SATELLITE

CSID	LOAD DTS	NAME	RCRD SRC
0	10-12-2000	N/A	SYSTEM
1	10-12-2000	ABC Suppliers	MANUFACT
1	10-14-2000	ABC Suppliers, Inc	MANUFACT
1	10-31-2000	ABC Worldwide Suppliers, Inc	MANUFACT
1	12-2-2000	ABC DEF Incorporated	CONTRACTS
2	10-2-2000	WorldPart	CONTRACTS
2	10-14-2000	Worldwide Suppliers Inc	CONTRACTS
3	***10-1-2000***	***N/A***	***FINANCE***

Dummy satellite record eliminates need for outer joins during extract.

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Satellite Entity – Link Related

Link Order Details

ID	Product ID	OrdID	LOAD DTS	RCRD SRC
0	0	0	10-12-2000	SYSTEM
1	PRD102	1	10-12-2000	MANUFACT
2	PRD103	1	10-2-2000	CONTRACTS

Satellite Order Totals

ID	LOAD DTS	Tax	Total	RCRD SRC
0	10-12-2000	<NULL>	<NULL>	SYSTEM
1	10-12-2000	3.00	0.00	MANUFACT
1	10-14-2000	4.00	12.00	MANUFACT
1	10-31-2000	3.69	14.02	MANUFACT
1	12-2-2000	4.69	13.69	CONTRACTS
2	10-2-2000	2.45	10.00	CONTRACTS
2	10-14-2000	1.22	14.00	CONTRACTS

Dummy satellite record eliminates need for outer joins during extract.

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Satellite Splits – Type of Information

Hub Customer

ID	CUSTOMER #	LOAD DTS	RCRD SRC
0	N/A	10-12-2000	SYSTEM
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-2-2000	CONTRACTS
3	*ABC5525-25*	***10-1-2000***	***FINANCE***

CUSTOMER SATELLITE

CSID	LOAD DTS	NAME	Contact	Sales Rgn	Cust Score	RCRD SRC
0	10-12-2000	N/A	N/A	N/A	0	SYSTEM
1	10-12-2000	ABC Suppliers	Jen F.	SE	102	MANUFACT
1	10-14-2000	ABC Suppliers, Inc	Jen F.	SE	120	MANUFACT
1	10-31-2000	ABC Worldwide Suppliers, Inc	Jen F.	SE	130	MANUFACT
1	12-2-2000	ABC DEF Incorporated	Jack J.	SC	85	CONTRACTS
2	10-2-2000	WorldPart	Jenny	SE	99	CONTRACTS
2	10-14-2000	Worldwide Suppliers Inc	Jenny	SE	102	CONTRACTS
3	***10-1-2000***	***N/A***	***N/A***	***N/A***	***0***	***FINANCE***

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Satellite Splits – Type of Information

Hub Customer

ID	CUSTOMER #	LOAD DTS	RCRD SRC
0	N/A	10-12-2000	SYSTEM
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-2-2000	CONTRACTS
3	*ABC5525-25*	***10-1-2000***	***FINANCE***

Customer Name Satellite (name Info)

Customer Sales Satellite (Sales Info)

- Because of the type of information is different, we split the logical groups into multiple Satellites.
- This provides sheer flexibility in representation of the information.
- We may have one more problem with Rate Of Change…

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Satellite Splits – Rate of Change

Hub Customer

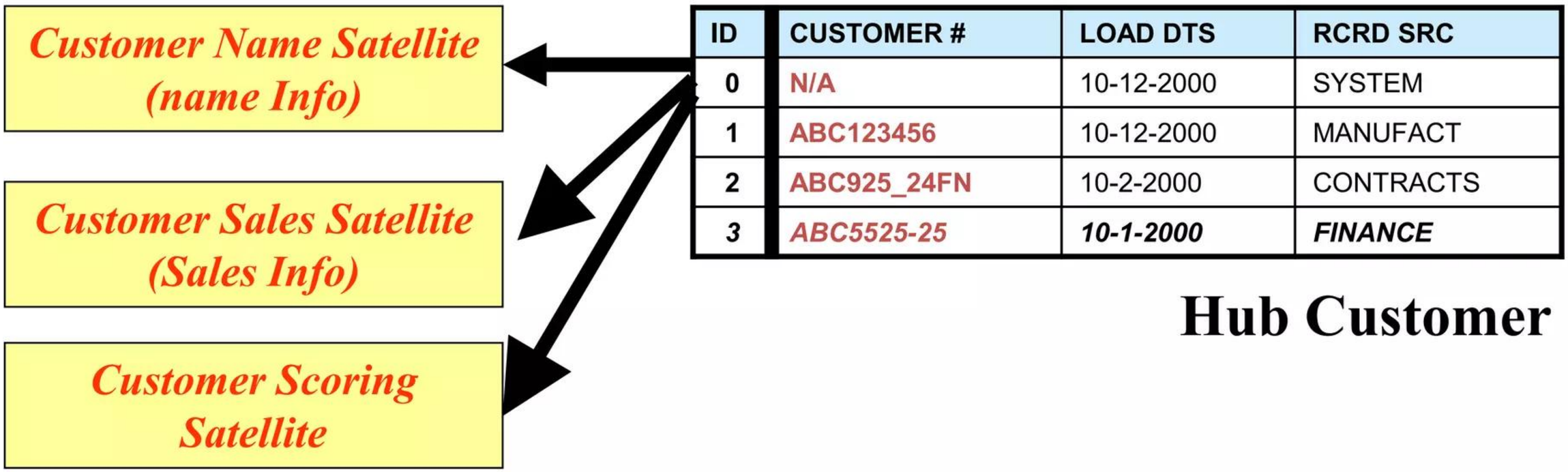
ID	CUSTOMER #	LOAD DTS	RCRD SRC
0	N/A	10-12-2000	SYSTEM
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-2-2000	CONTRACTS
3	*ABC5525-25*	***10-1-2000***	***FINANCE***

CUSTOMER SATELLITE

CSID	LOAD DTS	NAME	Contact	Sales Rgn	Cust Score	RCRD SRC
0	10-12-2000	N/A	N/A	N/A	0	SYSTEM
1	10-12-2000	ABC Suppliers	Jen F.	SE	102	MANUFACT
1	10-14-2000	ABC Suppliers, Inc	Jen F.	SE	120	MANUFACT
1	10-31-2000	ABC Worldwide Suppliers, Inc	Jen F.	SE	130	MANUFACT
1	12-2-2000	ABC DEF Incorporated	Jack J.	SC	85	CONTRACTS
2	10-2-2000	WorldPart	Jenny	SE	99	CONTRACTS
2	10-14-2000	Worldwide Suppliers Inc	Jenny	SE	102	CONTRACTS
3	***10-1-2000***	***N/A***	***N/A***	***N/A***	***0***	***FINANCE***

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Satellite Splits – Rate of Change



ID	CUSTOMER #	LOAD DTS	RCRD SRC
0	N/A	10-12-2000	SYSTEM
1	ABC123456	10-12-2000	MANUFACT
2	ABC925_24FN	10-2-2000	CONTRACTS
3	*ABC5525-25*	***10-1-2000***	***FINANCE***

Hub Customer

- Assume the data to score customers begins arriving in the warehouse every 5 minutes… We then separate the scoring information from the rest of the satellites.
- IF we end up with data that (over time) doesn't change as much as we thought, we can *always* re-combine Satellites to eliminate joins.

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Satellites Split By Source System

SAT_SALES_CUST

PARENT SEQUENCE
LOAD DATE
<LOAD-END-DATE>
<RECORD-SOURCE>
Name
Phone Number
Best time of day to reach
Do Not Call Flag

SAT_FINANCE_CUST

PARENT SEQUENCE
LOAD DATE
<LOAD-END-DATE>
<RECORD-SOURCE>
First Name
Last Name
Guardian Full Name
Co-Signer Full Name
Phone Number
Address
City
State/Province
Zip Code

SAT_CONTRACTS_CUST

PARENT SEQUENCE
LOAD DATE
<LOAD-END-DATE>
<RECORD-SOURCE>
Contact Name
Contact Email
Contact Phone Number

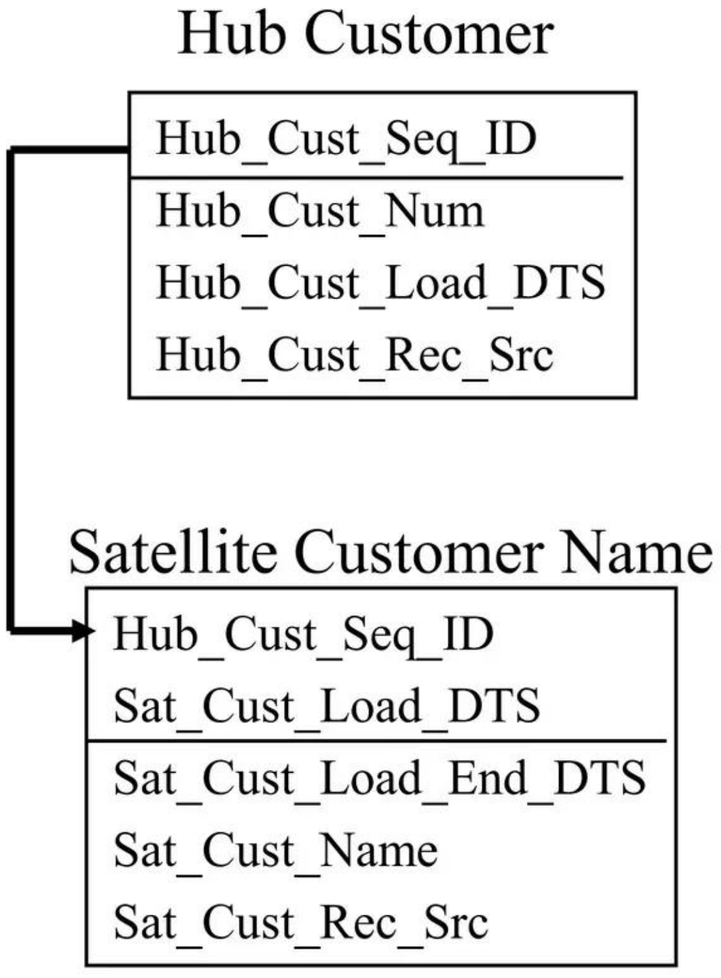
Satellite Structure

PARENT SEQUENCE
LOAD DATE
<LOAD-END-DATE>
<RECORD-SOURCE>
{user defined descriptive data}
{or temporal based timelines}

Primary Key

What Is the World's Smallest Data Vault?

Worlds Smallest Data Vault



- The Data Vault doesn't have to be "BIG".
- An Data Vault can be built incrementally.
- Reverse engineering one component of the existing models is not uncommon.
- Building one part of the Data Vault, then changing the marts to feed from that vault is a best practice.

- The smallest Enterprise Data Warehouse consists of two tables:
  - One Hub,
  - One Satellite

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Top 10 Rules for DV Modeling

Business keys with a low propensity for change become Hub keys.

Transactions and integrated keys become Link tables.

Descriptive data always fits in a Satellite.

1. A Hub table always migrates its' primary key outwards.
2. Hub to Hub relationships are allowed only through a link structure.
3. Recursive relationships are resolved through a link table.
4. A Link structure must have at least 2 FK relationships.
5. A Link structure can have a surrogate key representation.
6. A Link structure has no limit to the number of hubs it integrates.
7. A Link to Link relationship is allowed.
8. A Satellite can be dependent on a link table.
9. A Satellite can only have one parent table.
10. A Satellite cannot have any foreign key relationships except the primary key to the parent table (hub or link).

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NOTE: Automating the Build

- DV is a repeatable methodology with rules and standards
- Standard templates exist for:
  - Loading DV tables
  - Extracting data from DV tables
- RapidAce (www.rapidace.com – now Open Source)
  - Software that applies these rules to:
    - Convert 3NF models to DV
    - Convert DV to Star Schema
- This could save us lots of time and $$

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In Review…

- Data Vault is…
  - A Data Warehouse Modeling Technique (& Methodology)
  - Hub and Spoke Design
  - Simple, Easy, Repeatable Structures
  - Comprised of Standards, Rules & Procedures
  - Made up of Ontological Metadata
  - ***AUTOMATABLE!!!***
- Hubs = Business Keys
- Links = Associations / Transactions
- Satellites = Descriptors

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The Experts Say…



"The Data Vault is the optimal choice for modeling the EDW in the DW 2.0 framework." Bill Inmon

"The Data Vault is foundationally strong and exceptionally scalable architecture." Stephen Brobst





"The Data Vault is a technique which some industry experts have predicted may spark a revolution as the next big thing in data modeling for enterprise warehousing...." Doug Laney

More Notables…



"This enables organizations to take control of their data warehousing destiny, supporting better and more relevant data warehouses in less time than before." Howard Dresner

"[The Data Vault] captures a practical body of knowledge for data warehouse development which both agile and traditional practitioners will benefit from.." Scott Ambler



Who's Using It?

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Qwest

FEDERAL AVIATION ADMINISTRATION

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ABN·AMRO

DEPARTMENT OF THE NAVY · UNITED STATES OF AMERICA

StudentLoanXpress® A CIT Company

IHS

SNS Bank

UNITED STATES ARMY

DAIMLER

Tyson

TELE2

logica

BlueCross BlueShield

Mellon

CIGNA

Microsoft

DEPARTMENT OF DEFENSE · UNITED STATES OF AMERICA

Statistics Netherlands

MERCY HEALTH SYSTEM
Member of the Mercy Alliance, Inc.

nutreco

JPMorganChase

Growing Adoption…

- The number of Data Vault users in the US surpassed **500** in 2010 and grows rapidly (http://danlinstedt.com/about/dv-customers/)

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Conclusion?



Changing the direction of the river takes less effort than stopping the flow of water

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Super Charge Your Data Warehouse

Invaluable Data Modeling Rules to Implement Your Data Vault

Dan Linstedt

Inventor of the Data Vault

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Where To Learn More

The Technical Modeling Book: http://LearnDataVault.com

On YouTube: http://www.youtube.com/LearnDataVault

On Facebook: www.facebook.com/learndatavault

Dan's Blog: www.danlinstedt.com

The Discussion Forums: http://LinkedIn.com – Data Vault Discussions

World wide User Group (Free): http://dvusergroup.com

The Business of Data Vault Modeling
by Dan Linstedt, Kent Graziano, Hans Hultgren
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THANKS FOR ATTENDING!